Simi Shrivastava

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TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, SciPy, scikit-learn, PyTorch, Matplotlib), SQL (Window Functions, Joins, CTEs), R, Java, C/C++
- **Tools & Platforms:** Excel (Pivot Tables, VLOOKUP/XLOOKUP), Tableau, Power BI, Snowflake, Git, Docker, Apache Spark, ClickHouse, AWS (EC2, S3, RDS, Lambda, IAM), Jenkins, JIRA
- **Analytics:** Statistical Modeling, EDA, Hypothesis Testing, A/B Testing, Predictive Modeling, Data Visualization, Feature Engineering, Time-series Analysis
- **Machine Learning:** Classification, Regression, Clustering, Ensemble Methods (Random Forest, XGBoost), NLP, Deep Learning, Hyperparameter Tuning, Cross-validation

WORK EXPERIENCE

Data Science Intern | Canaria, Remote, US Jun 2025 – Aug 2025

- Analyzed and aggregated 20M+ records from 15+ data sources using SQL joins and window functions, building a production-grade salary intelligence pipeline for enterprise-scale compensation benchmarking.
- Integrated geographic cost-of-living adjustment algorithms using U.S. Census, BLS, and county-level economic data, combining internal and external metrics to improve regional accuracy of salary insights.
- Developed automated anomaly detection framework using statistical thresholds and BLS cross-validation, implementing data quality monitoring and validation pipelines that cut manual review effort by 60%.
- Collaborated cross-functionally with data engineers to build ETL workflows using Apache Spark, ClickHouse, and Parquet, reducing end-to-end data processing time by 70%.

Data Analyst | Cognizant Technology Solutions, India Nov 2019 – Jul 2024

- Performed statistical analysis and predictive modeling on financial datasets using Python (Pandas, NumPy), SQL, and Excel pivot tables, improving market trend prediction accuracy by 20% and enabling stock exchange clients to make more informed index rebalancing decisions.
- Presented analytical findings to stock exchange clients after each monthly release and built dashboards using Tableau, Power BI, and Excel (pivot tables, charts) that enabled stakeholders to identify data quality issues early and track index rebalancing outcomes across multiple client engagements.
- Trained and validated machine learning classification models (KNN, Decision Trees) for automated bug detection, implementing evaluation workflows with confusion matrices and cross-validation that reduced manual validation time from 1,500 to 4 minutes per 100 test cases, generating $14K in monthly cost savings.
- Designed data validation tool for financial system migration, consolidating data across sources using Excel (VLOOKUP/XLOOKUP) and SQL, reducing reconciliation time from 8 hours to 5 minutes while mentoring junior analysts on SQL, Python, and analysis workflows.

EDUCATION

University of Maryland, College Park, MD, US Aug 2024 – May 2026

Master of Science in Data Science, GPA: 3.84/4.0

Relevant Coursework: Machine Learning, Algorithms for Data Science, Big Data Systems, Natural Language Processing, Cloud Computing, Deep Learning

CSVTU, India Jul 2015 – May 2019

Bachelor of Engineering in Electronics and Telecommunication, GPA: 8.53/10.00

ACADEMIC PROJECTS

YOLOv8 Object Detection via Scalable AWS Cloud Pipeline

- Deployed a production-grade YOLOv8 model on AWS (EC2, S3, ECR, CloudWatch, IAM), processing 5,000+ images with Auto Scaling Groups and Application Load Balancer to establish cost-per-inference and throughput benchmarks.
- Containerized the inference API with Docker and automated deployment via ECR, reducing environment provisioning time by 60% and building an end-to-end ML pipeline with real-time monitoring.

Sentiment Classification using RNN Architectures

- Implemented and compared 36 configurations of RNN, LSTM, and Bidirectional LSTM architectures for binary sentiment classification on the IMDB dataset (50,000 reviews), evaluating optimizers, activation functions, sequence lengths, and gradient clipping using PyTorch and NLTK.
- Achieved 81.6% accuracy with the optimal LSTM configuration, demonstrating that optimizer selection (Adam vs. SGD) had a greater impact on classification performance than architectural complexity.

Ocular Disease Recognition using Machine Learning

- Preprocessed and analyzed 10,000+ ocular fundus images, performing EDA, statistical hypothesis testing, and feature analysis to uncover correlations between age, lifestyle factors, and disease prevalence.
- Achieved 91% classification accuracy by implementing and optimizing CNN models for multi-class disease detection with hyperparameter tuning, regularization, and model evaluation using confusion matrices and ROC curves.